PROJECT REPORT

Project Title

Student Attendance and Academic Performance Tracker Using Zoho Creator

1. Project Description

The Student Attendance and Academic Performance Tracker is a low-code application developed using Zoho Creator. It helps educational institutions maintain student details, record daily attendance, manage academic marks, and generate performance reports.

The application uses Zoho Creator forms, reports, dashboards, workflows, and Deluge scripting to automate routine academic activities. It reduces manual paperwork, improves data organization, and helps faculty identify students with low attendance or poor academic performance.

2. Objectives

1. To maintain student information in a centralized database.
2. To record and manage daily student attendance.
3. To store assignment, test, and final examination marks.
4. To calculate attendance percentages automatically.
5. To identify students with low attendance and poor academic performance.
6. To generate reports and visual dashboards.
7. To automate notifications using workflows.
8. To explore AI-based academic performance analysis.

3. Forms Used

3.1 Student Details Form

This form stores basic student information.

Fields:

- Student ID
- Student Name
- Department
- Year
- Section

- Email
- Phone Number
- Joining Date

3.2 Attendance Records Form

This form records daily attendance.

Fields:

- Student (Lookup)
- Attendance Date
- Attendance Status (Present, Absent, Leave)
- Remarks

3.3 Academic Performance Form

This form stores academic marks.

Fields:

- Student (Lookup)
- Subject
- Assignment Mark
- Test Mark
- Final Mark
- Exam Date

4. Dataset Used

A sample dataset containing five students is used to test the application.

Student ID	Student Name	Attendance %	Assignment	Test	Final
101	Arun	90	85	78	82
102	Priya	75	92	88	90
103	Karthik	60	55	42	48
104	Divya	85	80	84	86
105	Rahul	70	65	60	62

Note: These are sample values for demonstration. Attendance percentages should be calculated from the actual attendance records.

5. Reports Generated

5.1 Student Details Report

Displays student IDs, names, departments, years, and contact information.

5.2 Attendance Report

Displays daily attendance records and allows faculty to filter students by attendance status.

5.3 Academic Performance Report

Displays assignment marks, test marks, and final marks for each student.

5.4 Low Attendance Report

Identifies students whose attendance percentage is below the specified threshold, such as 75%.

5.5 Low Marks Report

Identifies students whose final marks are below the specified threshold, such as 40 out of 100.

5.6 Dashboard

Displays student information, attendance summaries, performance reports, and charts in a single interface.

6. Three Deluge Scripts

The following examples use Zoho Creator's Deluge scripting language. Form and field link names must match the actual application configuration.

Script 1: Calculate Attendance Percentage

Purpose: Calculate the attendance percentage using the number of present days and total recorded attendance days.

```
presentCount = 0;

totalCount = 0;


for each rec in Attendance_Records

{

   if(rec.Student == input.Student)
```

```
    {
        if(rec.Attendance_Status != "Leave")
        {
            totalCount = totalCount + 1;


            if(rec.Attendance_Status == "Present")
            {
                presentCount = presentCount + 1;
            }
        }
    }
}


if(totalCount > 0)
{
    percentage = (presentCount * 100.0) / totalCount;
    input.Attendance_Percentage = percentage;
}
else
{
    input.Attendance_Percentage = 0;
}
```

Implementation note: Create an Attendance_Percentage decimal field in the relevant student or summary form. Configure this script in an appropriate workflow and adjust the Student lookup comparison to match the app's actual field type. For a reliable implementation, update the summary whenever attendance records are added, edited, or deleted.

Script 2: Identify Students with Low Attendance

Purpose: Flag students whose attendance is below 75%.

```
if(input.Attendance_Percentage < 75)
{
   input.Attendance_Status = "Low Attendance";
}
else
{
   input.Attendance_Status = "Regular";
}
```

Implementation note: Create suitable fields such as Attendance_Percentage and Attendance_Status. Use this logic in the relevant form workflow, with field link names matching your application.

Script 3: Identify Students with Low Marks

Purpose: Identify students who score below 40 in the final examination.

```
if(input.Final_Mark < 40)
{
   input.Performance_Status = "Needs Improvement";
}
else
{
   input.Performance_Status = "Satisfactory";
}
```

Implementation note: Create a Performance_Status field in the Academic Performance form and execute this script when the form is submitted or when marks are updated.

7. Workflow Automation

The application uses automated workflows to reduce manual work.

Workflow 1: Attendance Monitoring

- Trigger: Attendance record is created or updated.
- Action: Update attendance summary and percentage.
- Result: Faculty can monitor attendance regularly.

Workflow 2: Low Attendance Alert

- Trigger: Attendance percentage is updated.
- Condition: Attendance percentage is below 75%.
- Action: Flag the student and optionally send an email notification.
- Result: Faculty can identify students who need attendance improvement.

Workflow 3: Low Marks Alert

- Trigger: Academic Performance form is submitted or updated.
- Condition: Final mark is below 40.
- Action: Update performance status and send a notification if configured.
- Result: Faculty can provide timely academic support.

8. AI Features

The initial version uses rule-based automation. The following AI features can be added as optional enhancements.

8.1 Academic Performance Prediction

An external machine learning model can analyze attendance, assignment marks, and test marks to estimate whether a student may need academic support.

8.2 At-Risk Student Identification

AI can analyze historical academic records to identify patterns associated with poor performance.

8.3 Personalized Recommendations

The system can suggest revision activities, practice tests, or attendance improvement plans based on a student's performance.

8.4 AI Integration

An external AI service can be integrated using an API and Zoho Creator connections or supported Deluge integration methods.

Note: These AI capabilities require an actual model or AI service integration. They are not automatically available just because the application is built in Zoho Creator.

9. Application Workflow

1. The administrator or faculty member enters student details.
2. Student records are stored in the application.

3. Faculty records daily attendance.
4. Attendance records are used to calculate attendance percentages.
5. Faculty enters assignment, test, and final marks.
6. Deluge scripts classify attendance and academic performance.
7. Workflows trigger alerts when configured conditions are met.
8. Reports display student attendance and academic results.
9. The dashboard presents summaries and charts for monitoring.

10. Implementation Steps

1. Open the Zoho Creator website and sign in.
2. Create a new application named Student Attendance and Academic Performance Tracker.
3. Create the Student Details form.
4. Create the Attendance Records form with a Student lookup field.
5. Create the Academic Performance form.
6. Add sample student and attendance records.
7. Create student, attendance, and performance reports.
8. Create the dashboard page and add reports and charts.
9. Configure Deluge scripts for attendance calculation and performance classification.
10. Create workflows for low attendance and low marks.
11. Test the forms, reports, calculations, and notifications.
12. Configure user permissions and publish or share the application as required.

11. Expected Output

The completed application is expected to provide:

- A centralized database of student details.
- Daily attendance records for each student.
- Academic marks stored subject-wise.
- Automatic attendance percentage calculations.
- Identification of students with low attendance.

- Identification of students requiring academic improvement.
- Reports and dashboard charts for faculty.
- Automated alerts through configured workflows.

For the sample dataset, Karthik's final mark of 48 is above the example threshold of 40, so the low-marks rule will classify the record as satisfactory. Students with attendance below 75% will be flagged when the attendance percentage is calculated correctly.

12. Future Enhancements

1. Integrate an AI-based academic performance prediction model.
2. Add parent notifications through supported messaging integrations.
3. Generate downloadable attendance and performance reports.
4. Add role-based access for administrators, faculty, and students.
5. Introduce timetable and examination scheduling modules.
6. Add subject-wise and semester-wise performance analytics.
7. Develop a mobile-friendly interface for convenient access.
8. Integrate with existing college management systems through APIs.

13. Conclusion

The Student Attendance and Academic Performance Tracker demonstrates how Zoho Creator can be used to develop an educational management application with minimal coding. The application centralizes student information, attendance records, and academic marks while providing reports, dashboards, and automated workflows.

The use of Deluge scripting enables basic calculations and rule-based alerts. With additional AI integration, the system can be extended to provide academic risk prediction and personalized learning recommendations. Overall, the project improves record management, reduces repetitive manual work, and supports data-driven academic monitoring.